

IBM HR Attrition Analysis – Who Is Leaving & Why

Team 266

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Who is leaving?

Attrition by department, job level,
job role, and age group

Why are they leaving?

Income, satisfaction, equity structure

What does it cost?

\$13.6M annual replacement cost
by department

Key findings & recommendations

Four data-driven actions HR can
take immediately



IBM HR Employee Attrition

1,470 employees
3 departments
35 variables

16.1%

Overall Attrition Rate

237 of 1,470
employees left

\$13.6M

Estimated Annual Cost

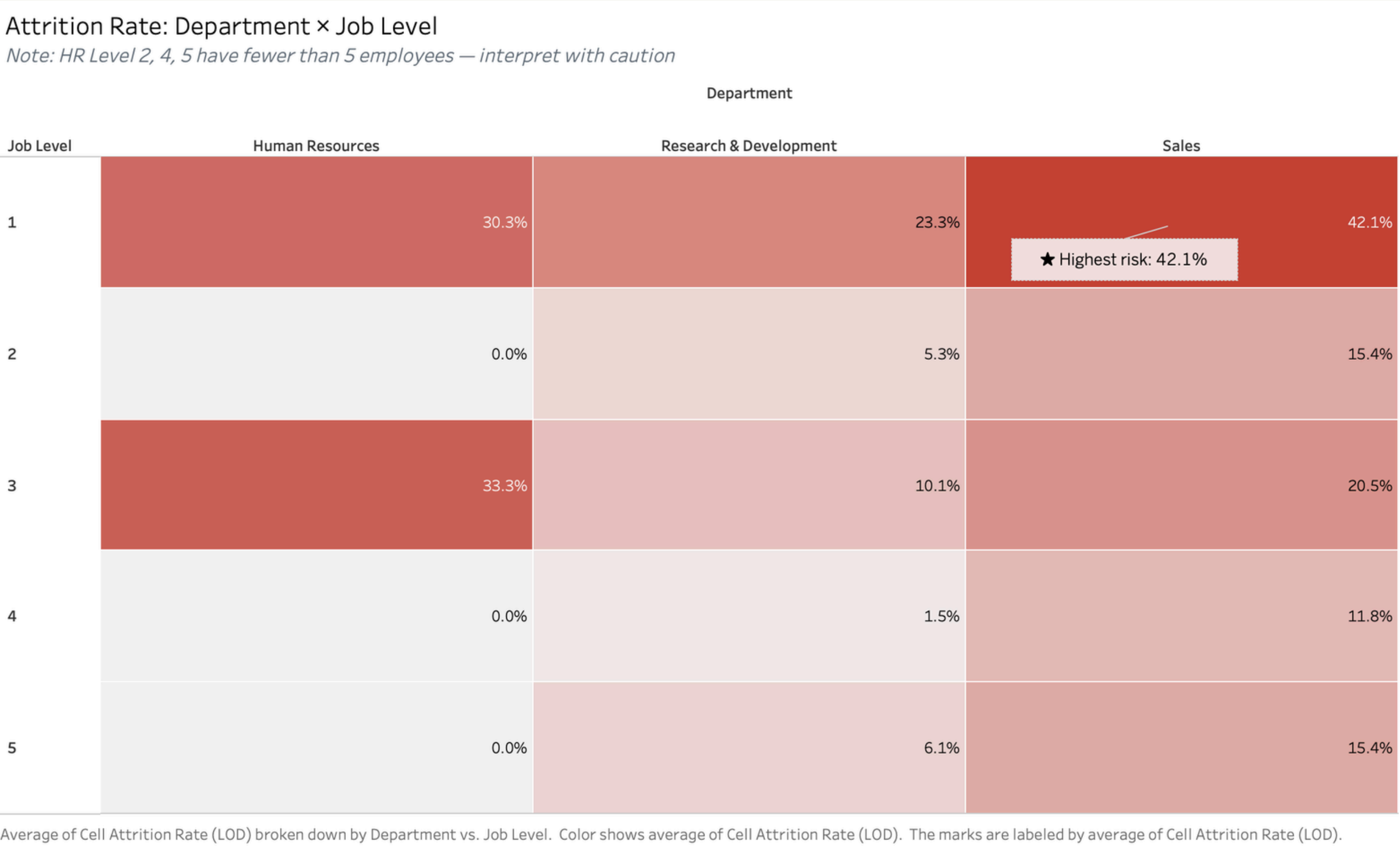
100% salary
replacement

42.1%

Peak Attrition Cell

Sales × Job
Level 1

Who Is Leaving? – Department × Job Level



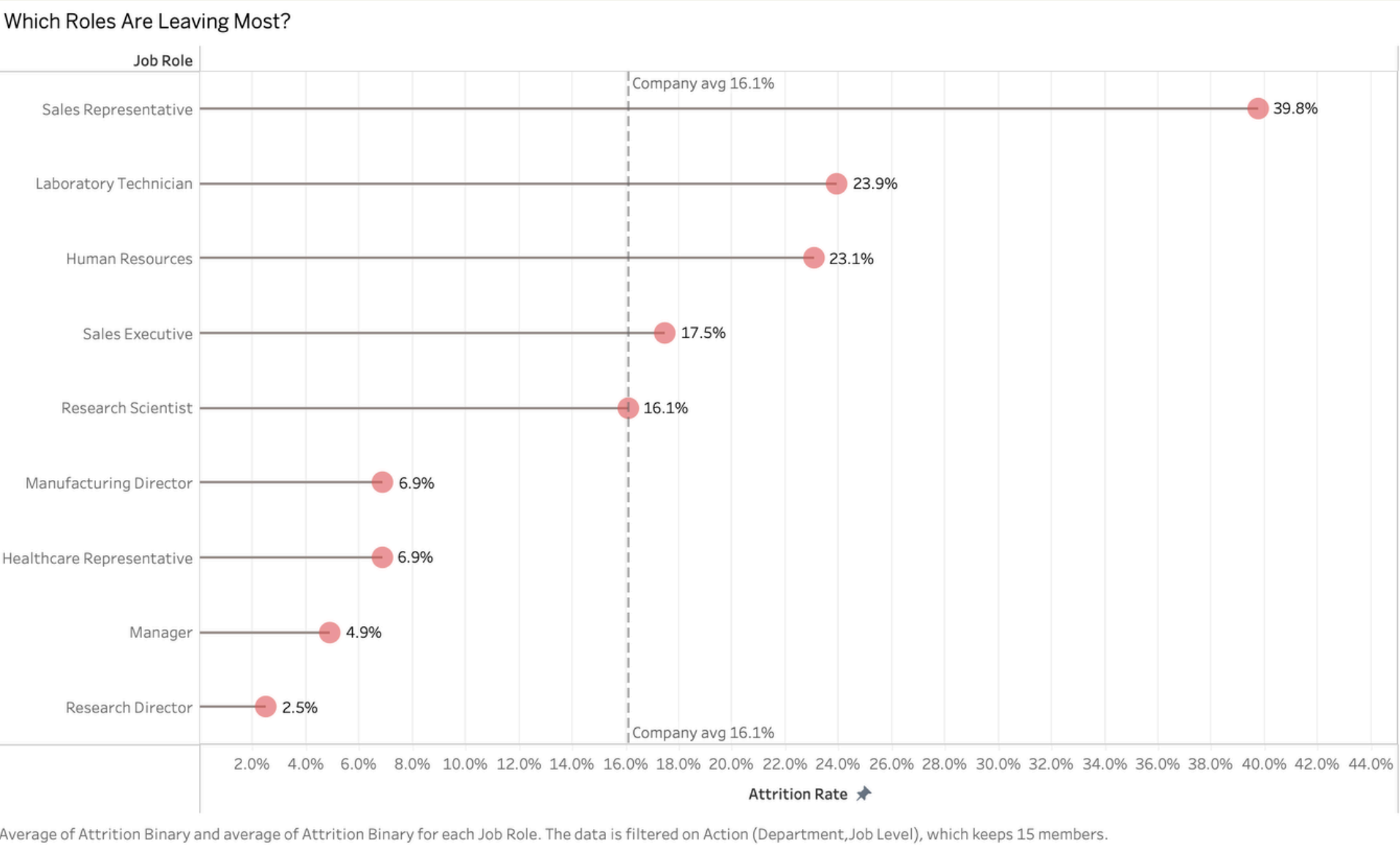
Key Findings:

- **Sales × Level 1 = 42.1%**
 - 2.6× company average
 - → Top priority for HR
- **HR Level 3 = 33.3%**
 - Small team (n<5), but still alarming
- **R&D Level 4 = 1.5%**
 - Safest zone in the entire company
- Level 1 employees across all departments are highest risk

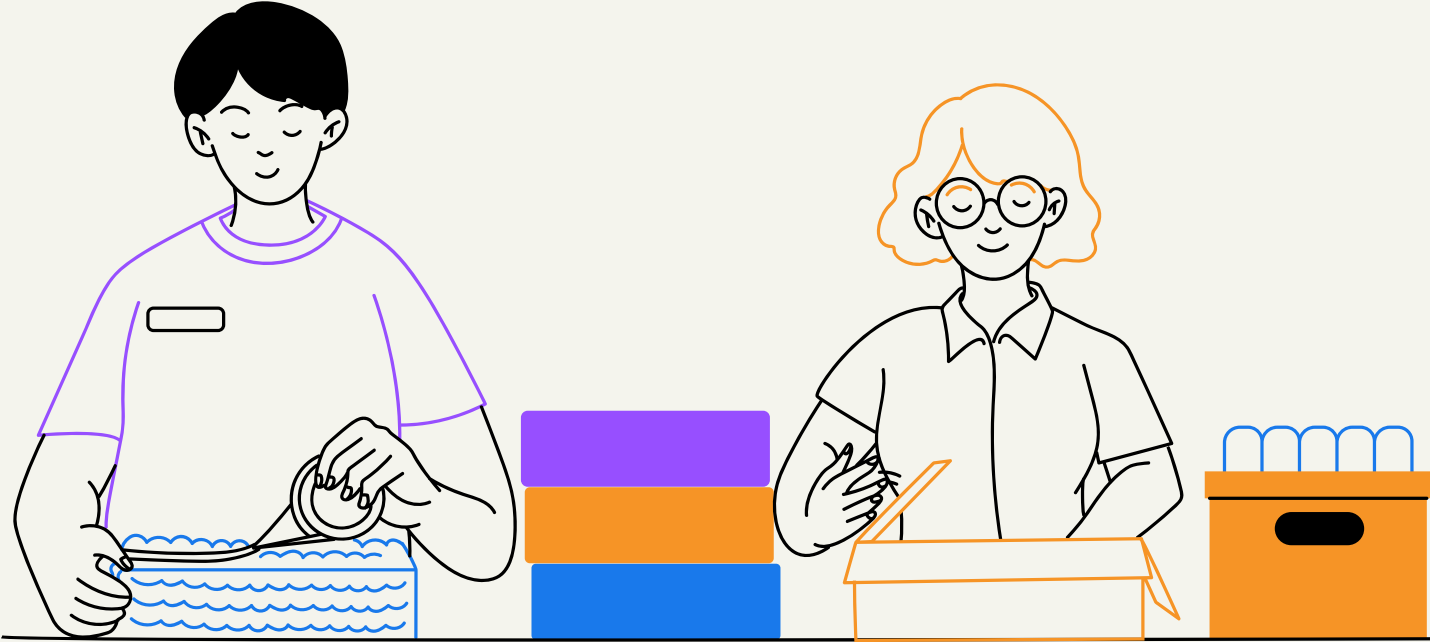
★ Sales × Level 1 has a 42.1% attrition rate — the highest-risk intersection. Every 2nd or 3rd entry-level Sales hire leaves. This is where HR intervention will have the highest ROI.



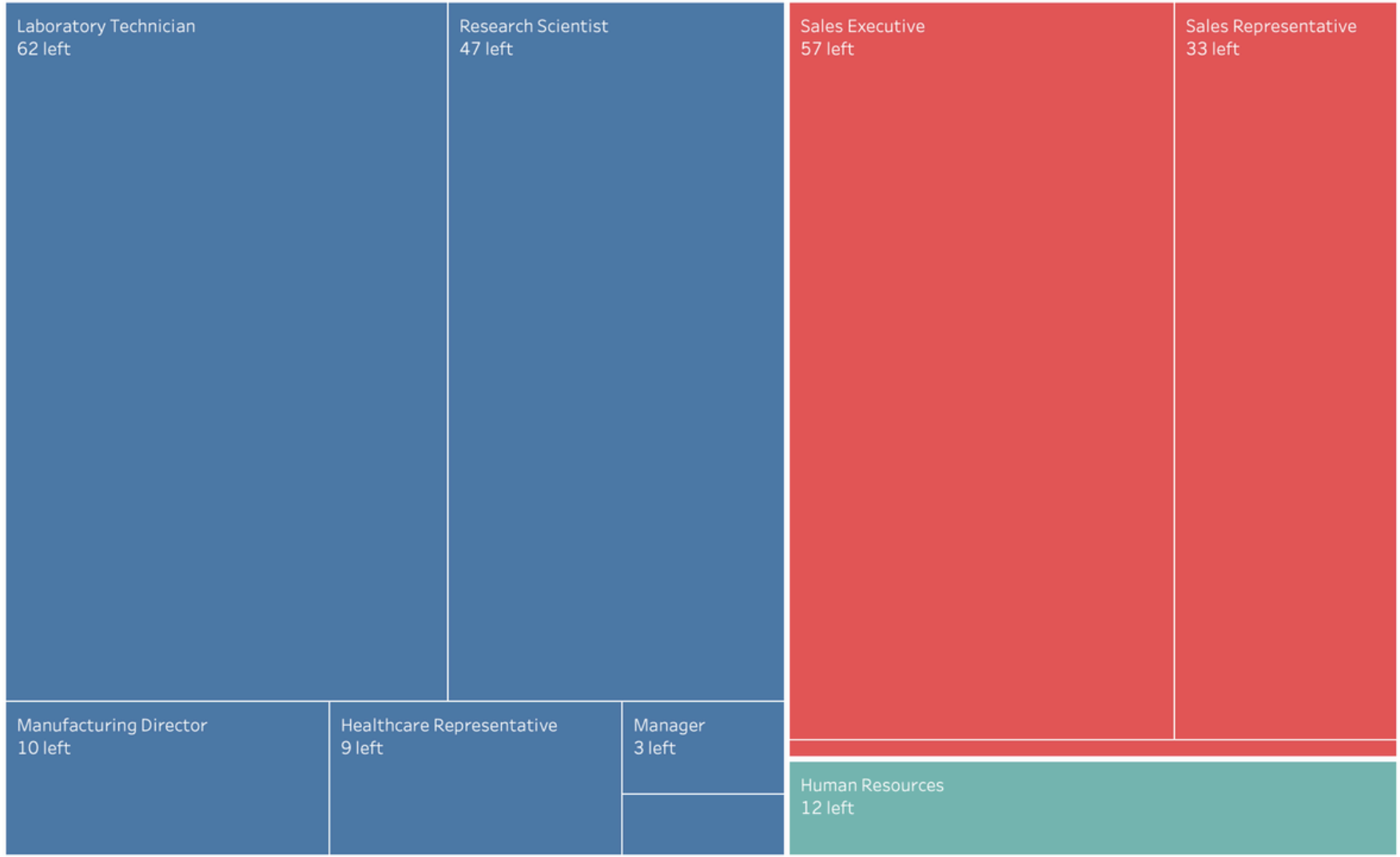
Which Roles Are Leaving Most?



- Rate vs. headcount tell different stories: Sales Rep has the highest attrition rate (39.8%) but only 33 people left.
- Lab Technicians are more stable (23.9%) yet contributed 62 departures — the most of any role.
- Research Director is the safest role at 2.5% — senior R&D staff are highly retained.
 - This pattern suggests seniority and specialization are protective factors.

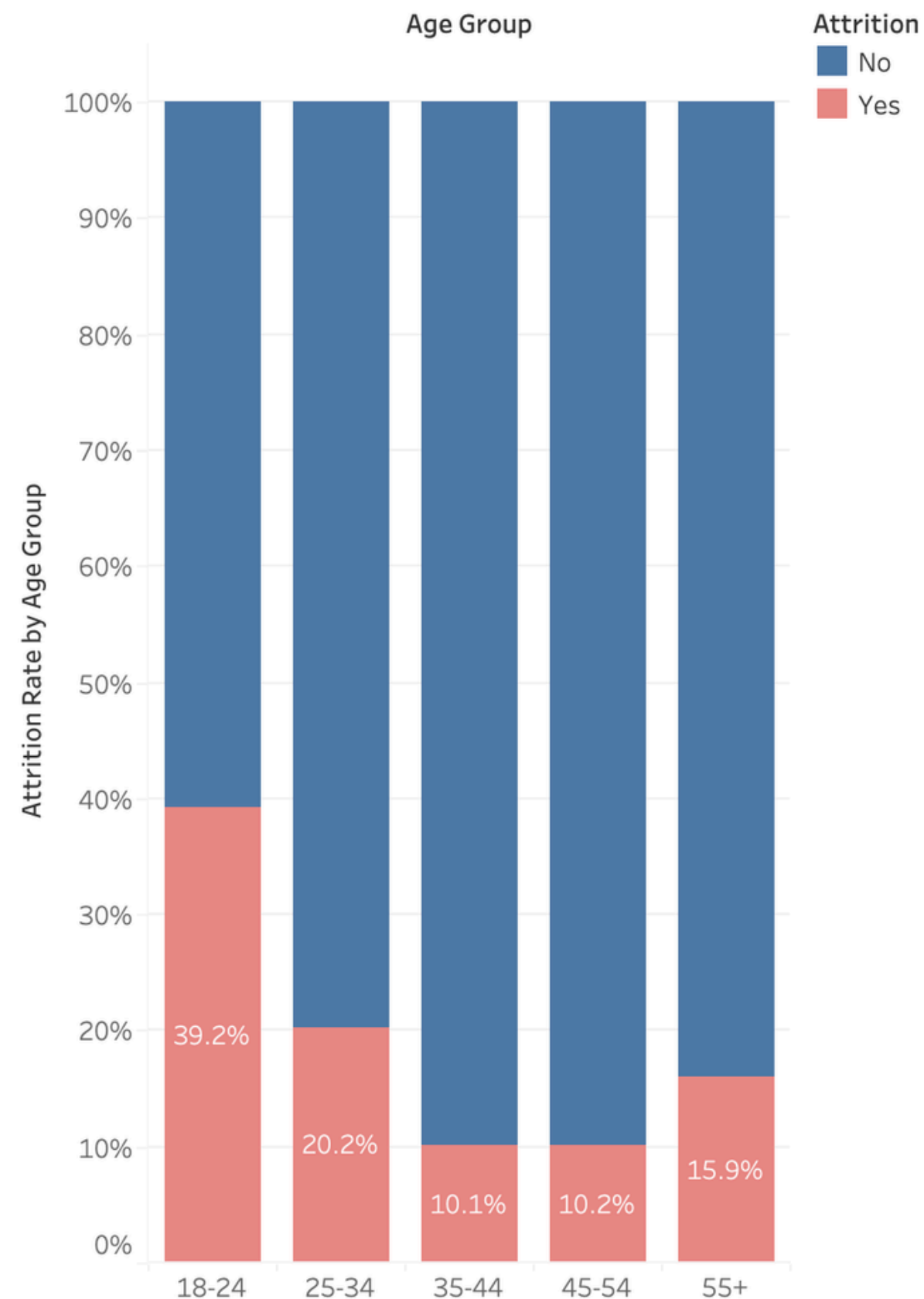


Who Left? Total Departures by Role



Job Role and count of Employee Number. Color shows details about Department. Size shows count of Employee Number. The marks are labeled by Job Role and count of Employee Number. Details are shown for Department. The data is filtered on Attrition and Action (Department,Job Level). The Attrition filter keeps Yes. The Action (Department,Job Level) filter keeps 15 members.

Attrition by Age Group

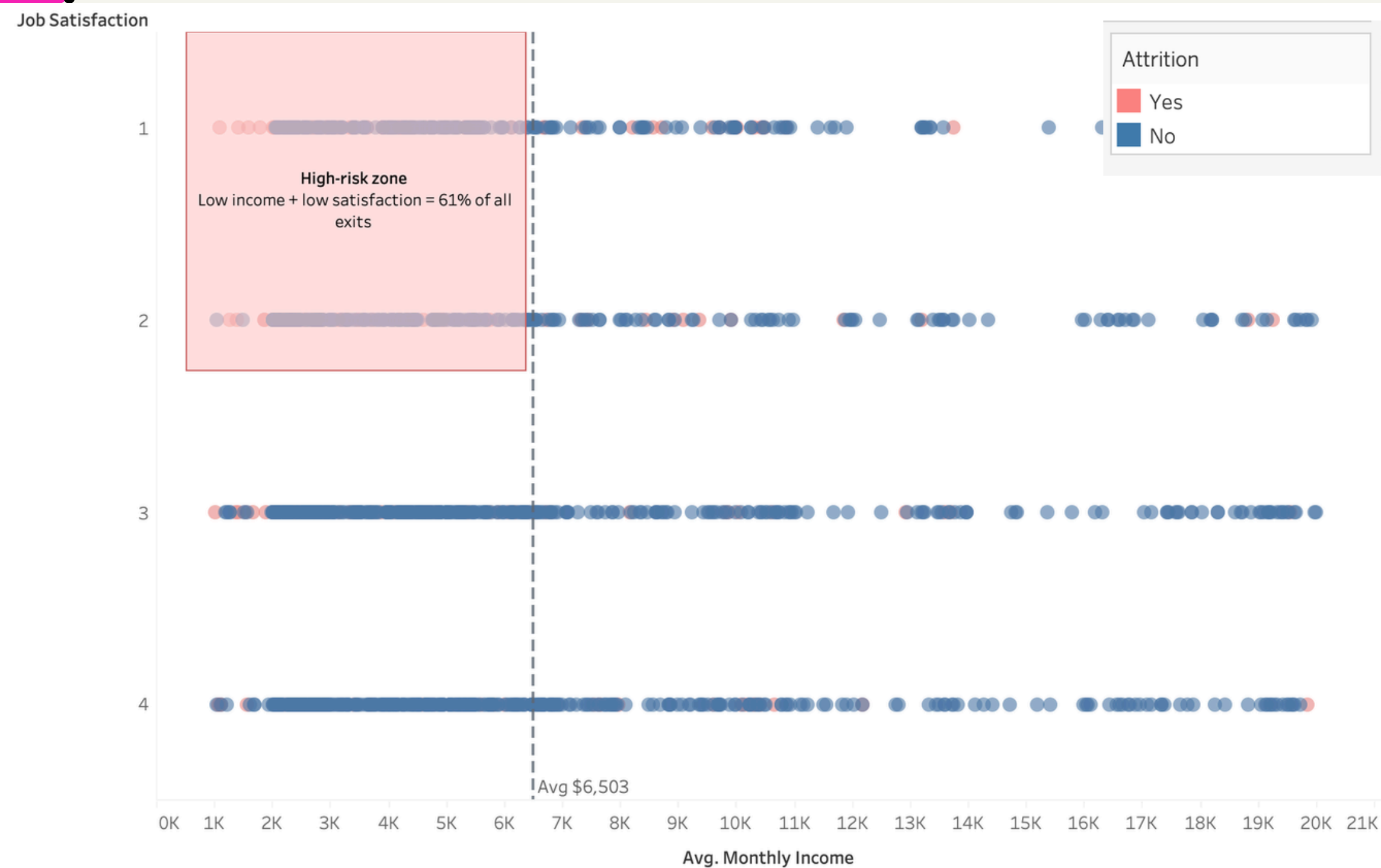


% of Total Count of Employee Number for each Age Group. Color shows details about Attrition.

Who Is Leaving? – By Age Group

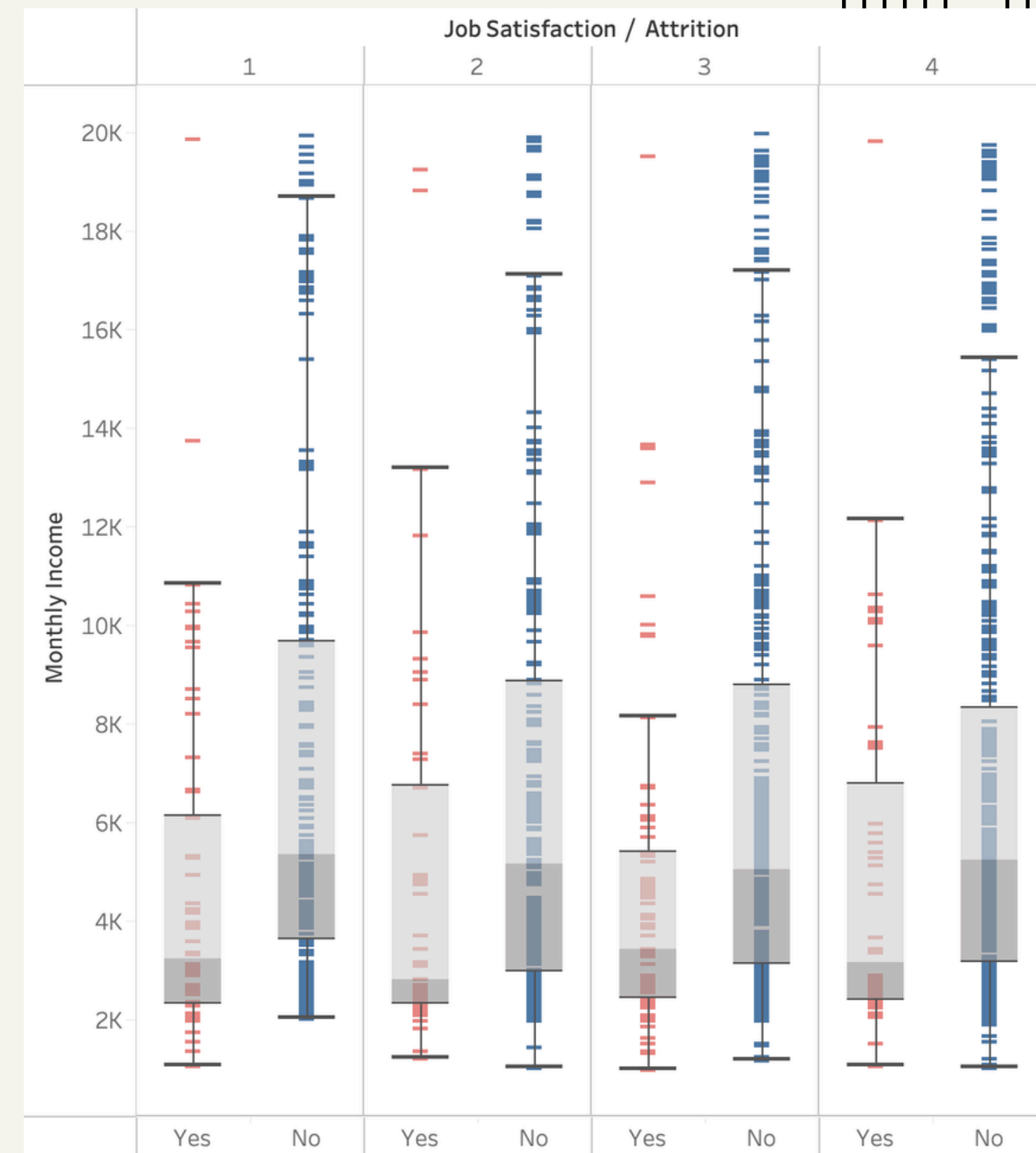
- **18–24: 39.2%**
 - Nearly 1 in 3 young employees leave
- **Sharp drop after 35**
 - Falls below 11% — mid-career stability
- **55+ uptick to 15.9%**
 - Possible pre-retirement or re-evaluation phase
- **25–34 is largest group (n=495)**
 - Improve retention here = biggest headcount impact
- Early-career retention is the company's largest demographic challenge.
- The 18–24 cohort loses nearly 2 in 5 employees.
- The 25–34 segment (n=495) has a 20.2% rate
 - Fix this age band would prevent the most departures in absolute terms.

Why Are They Leaving? – Income & Job Satisfaction

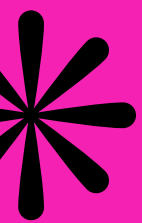


Average of Monthly Income for each Job Satisfaction. Color shows details about Attrition. Details are shown for various dimensions. The data is filtered on Action (Job Role), which keeps 9 members.

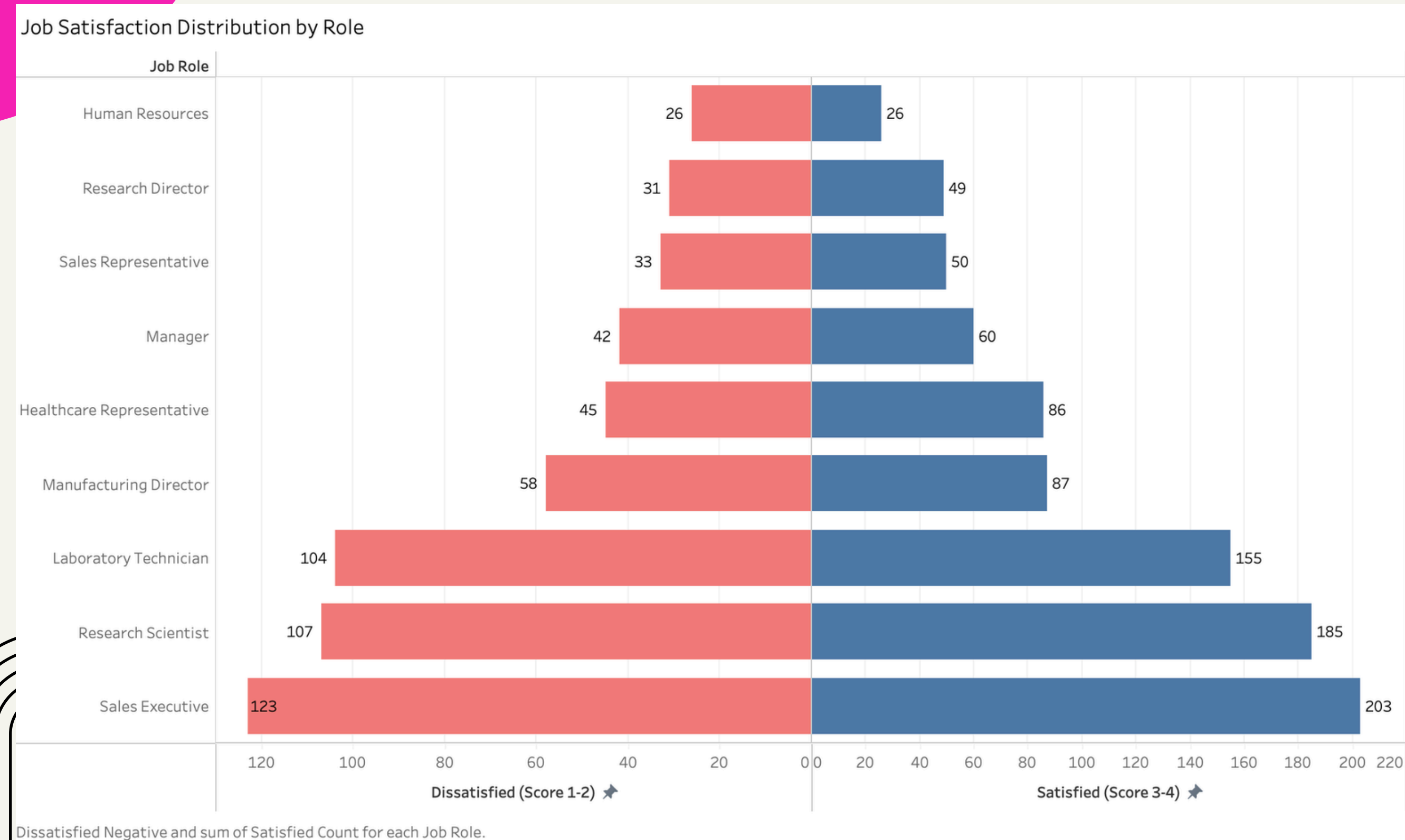
- –30% income gap: leavers avg \$4,787 vs stayers \$6,833
- Overtime attrition = 30.5% vs 10.4% non-overtime — 3× higher
- Single employees on overtime: 49.6% attrition — nearly 1 in 2
- Low income + low satisfaction quadrant = 61% of departures from 31% of headcount



Monthly Income for each Attrition broken down by Job Satisfaction. Color shows details about Attrition. Details are shown for Employee Number.



Why Are They Leaving? – Job Satisfaction Distribution by Role



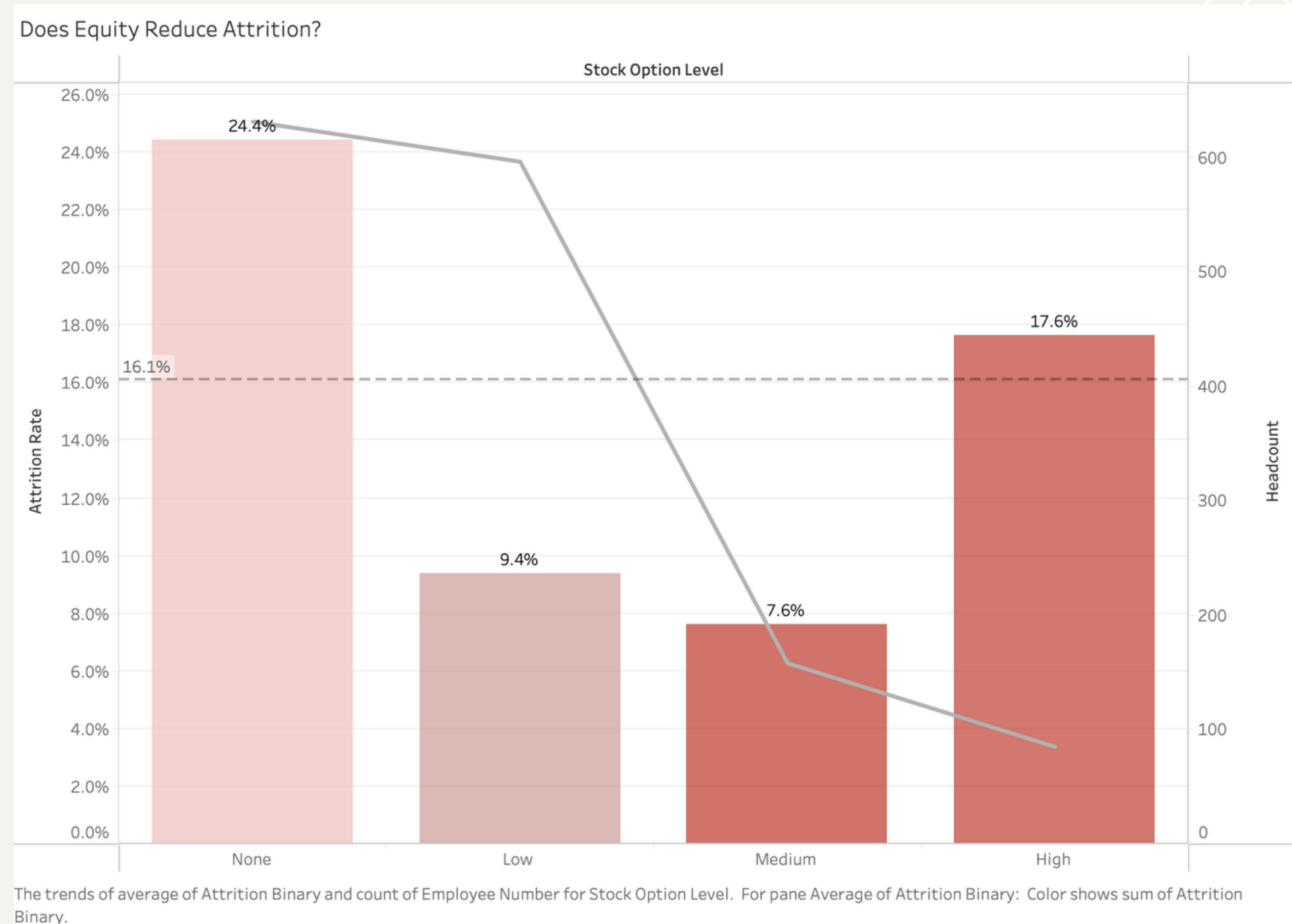
- **Sales Executive** has the most dissatisfied employees (123), and also the most satisfied (203).
 - Large polarized team. The dissatisfied group is a key retention risk.
- **Research Director** shows the best ratio: only 31 dissatisfied vs 49 satisfied.
 - Tracks with its 2.5% attrition rate.
- **Human Resources** has equal satisfied and dissatisfied counts (26 vs 26)
 - A 50/50 split suggesting structural issues within the team managing everyone else's wellbeing.

Why Are They Leaving? – Does Equity Reduce Attrition?

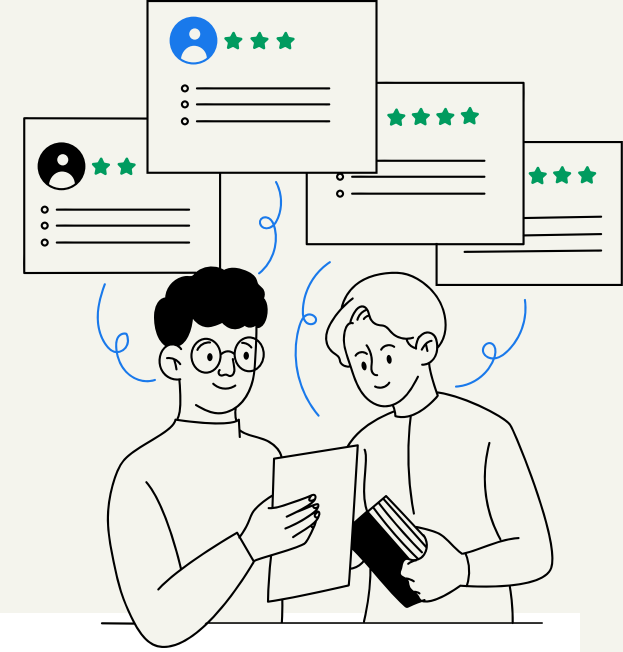
Note: Level 0 = None, Level 1 = Low, Level 2 = Medium, Level 3 = High

- Level 0 → Level 1: 24.4% → 9.4%
 - A 61% reduction in attrition rate
- Level 2 is lowest risk (7.6%) with 158 employees
 - Sweet spot for equity grants
- Level 3 rises to 17.6% but n=85 is too small for reliable inference
- Highest ROI action: move employees from Level 0 to Level 1

Even minimal equity grants (Level 1) cut attrition by 61%. This is the highest-ROI retention lever in the dataset. The cost of Level 1 grants is almost certainly lower than the \$13.6M annual attrition bill.



What Does It Cost? – Annual Attrition Replacement Cost



Assumptions

100% of annual salary per leaver
(industry conservative)

Adjustable parameter:

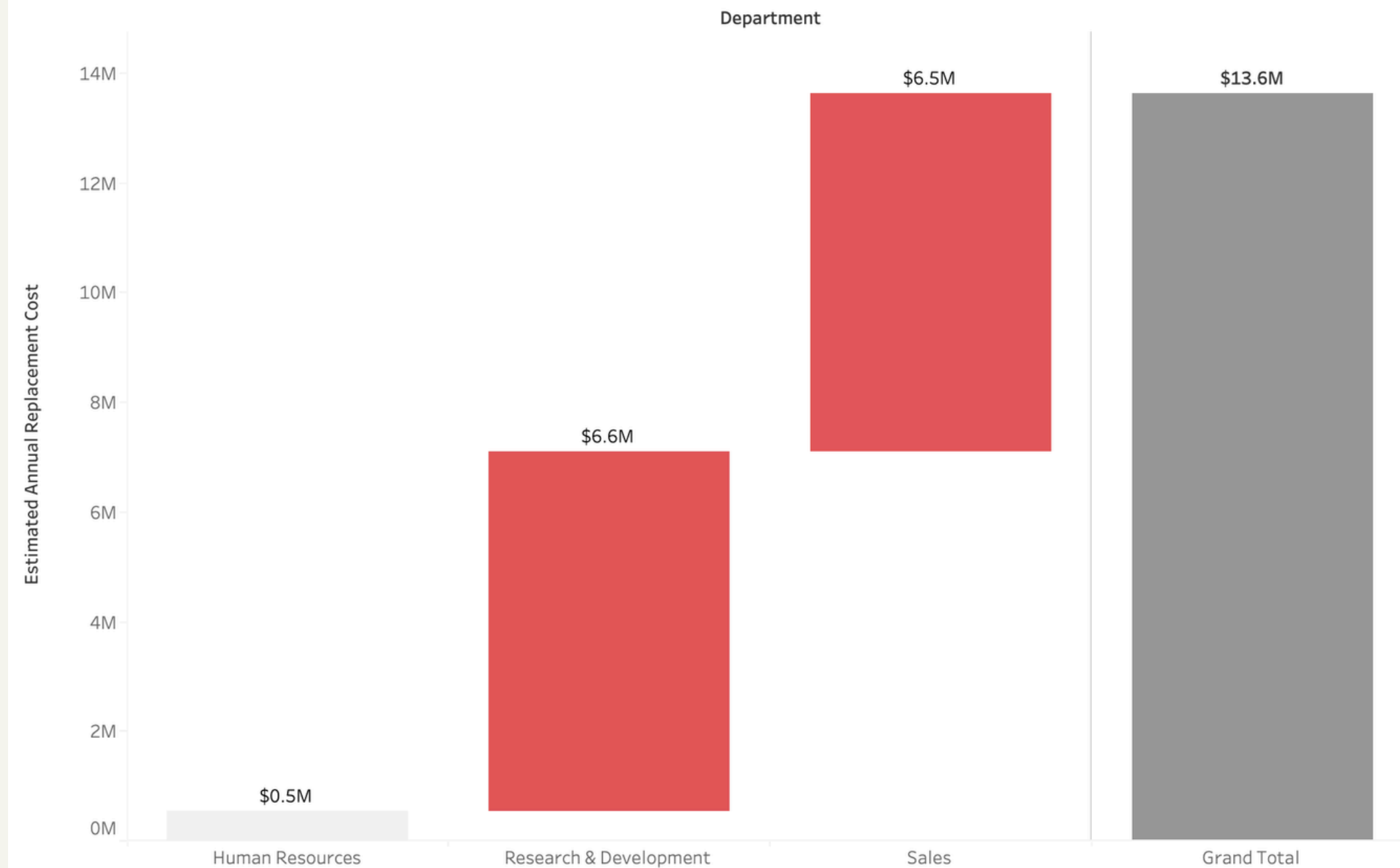
0.5× = \$6.8M

1.5× = \$20.4M

2.0× = \$27.2M

- R&D accounts for \$6.6M — largest share — due to high headcount (133 leavers at above-average salaries). Sales adds \$6.5M from 92 departures.
- \$13.6M is the conservative estimate. At 150% salary replacement, the figure rises to \$20.4M. This framing makes the business case for retention investment undeniable.

Estimated Annual Attrition Cost by Department



Gantt Start for each Department. Color shows sum of Attrition Cost. Size shows sum of Attrition Cost. The marks are labeled by sum of Attrition Cost.

01 – Intervene at Sales × Level 1 first

Finding: 42.1% attrition – 2.6× company average

Action: Structured mentorship, clear promotion path, and compensation review for entry-level Sales staff.

Why it matters: Every 2nd–3rd new hire in this group leaves. The cost of intervention is far lower than the cost of repeated replacement.

02 – Audit & reduce overtime for at-risk groups

Finding: OT employees leave at 30.5% vs 10.4% non-OT – 3× higher

Action: Identify employees with sustained overtime AND low satisfaction scores. Prioritize single employees on overtime – their attrition rate is 49.6%. Redistribute workload or hire additional headcount.

Why it matters: Policy change here could prevent ~65 exits per year – approximately \$2M in avoided replacement costs.

03 – Expand stock option access beyond Level 0

Finding: Level 0 → Level 1 cuts attrition from 24.4% to 9.4% (–61%)

Action: Identify all employees at Stock Level 0, especially in Sales and Lab Technician roles. Even minimal equity grants produce a measurable retention effect.

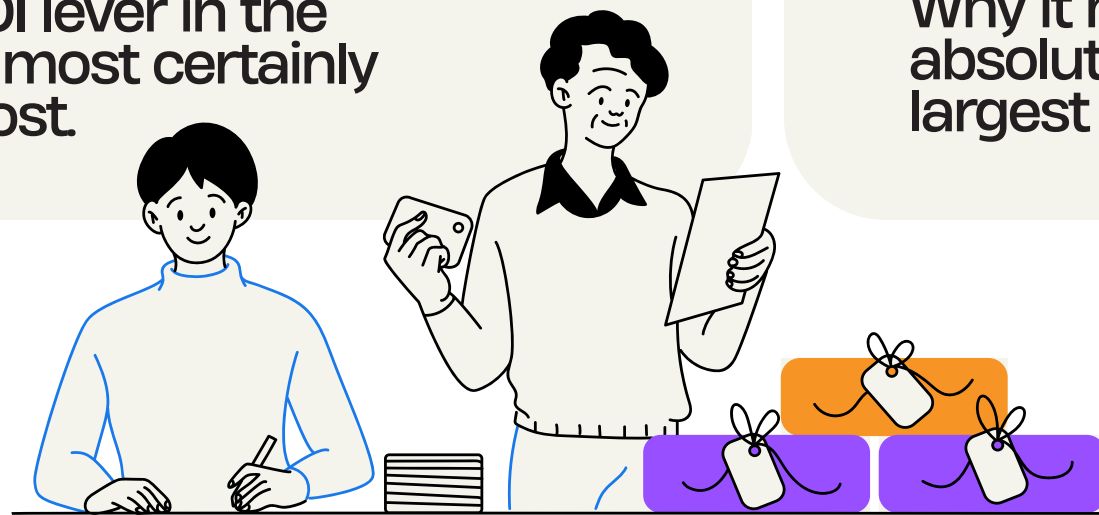
Why it matters: This is the highest-ROI lever in the dataset. The cost of Level 1 grants is almost certainly lower than the annual replacement cost.

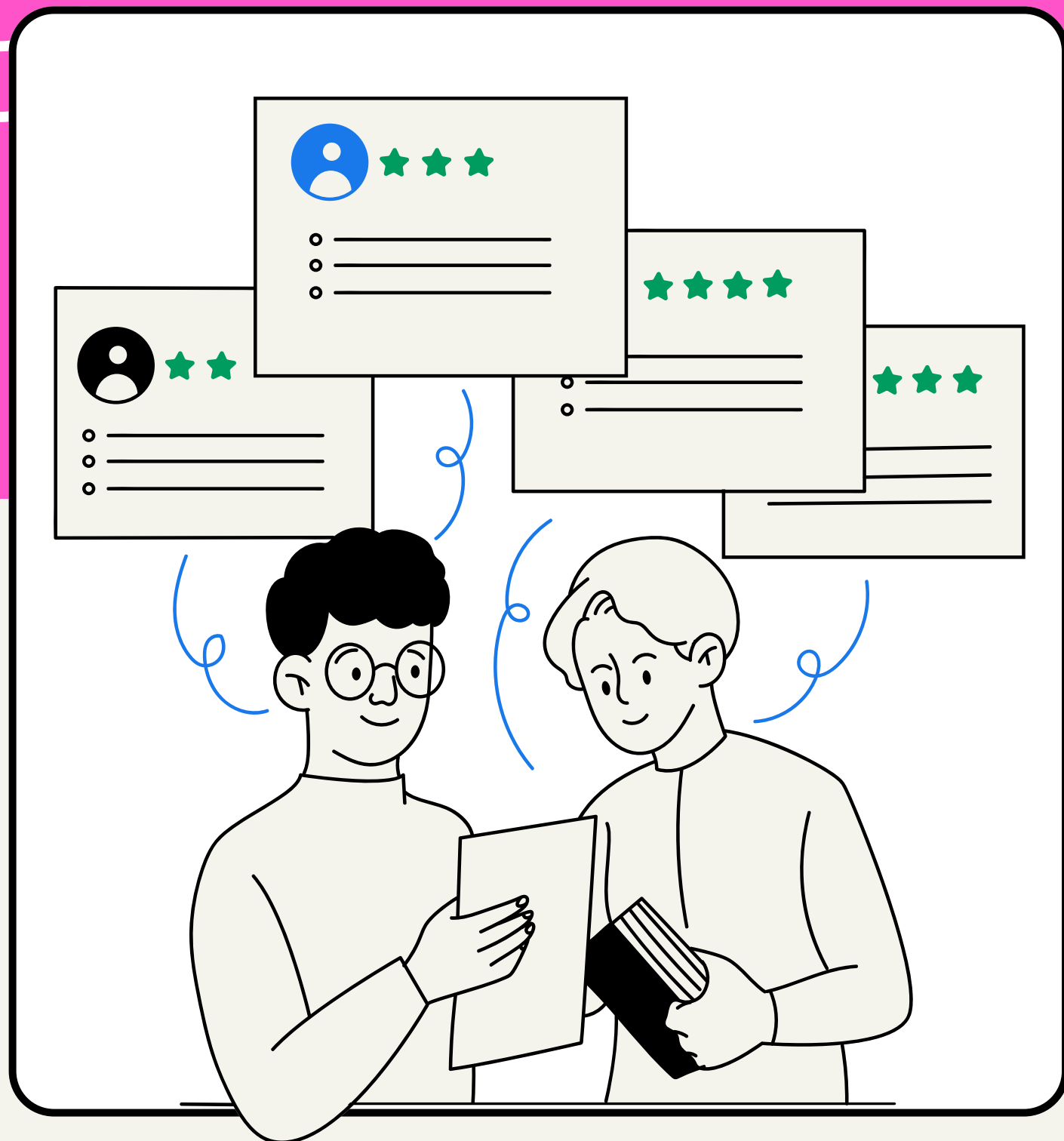
04 – Invest in early-career retention programs

Finding: 18–24 leave at 39.2%; 25–34 group has most exits in headcount

Action: Structured onboarding with 90-day check-ins, peer mentorship, and visible promotion timelines for employees under 30.

Why it matters: The 25–34 group generates the most absolute departures. Improving retention here has the largest headcount impact.





Thank You Very Much!

[Link to the Dashboard](#)

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